

Likelihood Ratio Test example

LRT set-up

Recall the coin-flipping scenario again! We have $X_1, \dots, X_n$ are a random sample from a $\text{Bern}(\theta)$ distribution, where θ is the probability of Heads. We have the hypotheses

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$$

We can conduct a LRT of these hypotheses. We found the likelihood ratio statistic to be:

$$\Lambda(\mathbf{X}) = \left(\frac{n\theta_0}{\sum X_i} \right)^{\sum X_i} \left(\frac{n(1-\theta_0)}{n - \sum X_i} \right)^{n - \sum X_i}$$

The LRT says reject when $\Lambda(\mathbf{X}) \leq k$ for some $k \in [0, 1]$. Notice that $\Lambda(\mathbf{X})$ depends on how many heads we see!

Specific case

Suppose we have $n = 6$ coin flips and $\theta_0 = 0.6$. That is,

$$H_0 : \theta = 0.6 \quad \text{vs.} \quad H_1 : \theta \neq 0.6$$

Also suppose we want a level-0.08 LRT of these hypotheses, which means we need to find the values of k such that

$$\Pr(\Lambda(\mathbf{X}) \leq k | \theta = 0.6) \leq 0.08$$

We can evaluate the test statistic $\Lambda(\mathbf{X})$ above for these specific values of n, θ_0 , and each possible value of $\sum_{i=1}^6 X_i = y \in \{0, 1, \dots, 6\}$:

```
n <- 6
theta0 <- 0.6
y <- 0:n
Lambda <- ((n*theta0/y)^y) * (n*(1-theta0)/(n - y))^(n - y)
```

Since $\Lambda(\mathbf{X})$ is a function of $\sum X_i$, we can obtain $\Pr(\Lambda(\mathbf{X}) = k | \theta = 0.6)$ from these Binomial PMF evaluations $\Pr(\sum X_i = y | \theta = 0.6)$. For example, $\Pr(\Lambda(\mathbf{X}) = 0.004096 | \theta = 0.6) = \Pr(\sum X_i = 0 | \theta = 0.6)$.

We display the possible values of $\Lambda(\mathbf{X})$ below, along with their associated probabilities (which are obtained from $\Pr(\sum X_i = y | \theta = 0.6)$):

	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$y = 5$	$y = 6$
$\Lambda(\mathbf{x})$	0.004	0.092	0.420	0.885	0.945	0.464	0.047
$Pr(\sum X_i = y \theta_0)$	0.004	0.037	0.138	0.276	0.311	0.187	0.047

Now we can find the value k such that $\Pr(\Lambda(\mathbf{X}) \leq k | \theta = 0.6) \leq 0.08$!

So rejecting H_0 when $\Lambda(\mathbf{X}) \leq k$ for any $k \in$ yields a level-0.08 test.

In particular, the test that rejects H_0 when $k =$ has size .